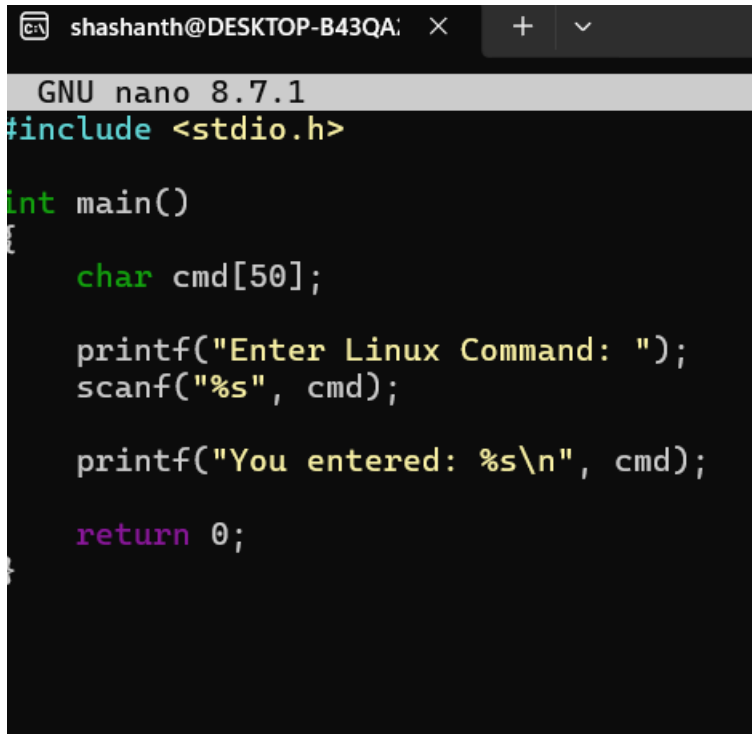


Name : M Shashanth

1.a

A screenshot of a terminal window with a nano editor. The title bar shows 'shashanth@DESKTOP-B43QA:'. The editor header says 'GNU nano 8.7.1'. The code is a C program that includes <stdio.h>, has a main function, declares a char array 'cmd' of size 50, prompts the user to 'Enter Linux Command:', reads the input with 'scanf', prints it back with 'printf', and returns 0.

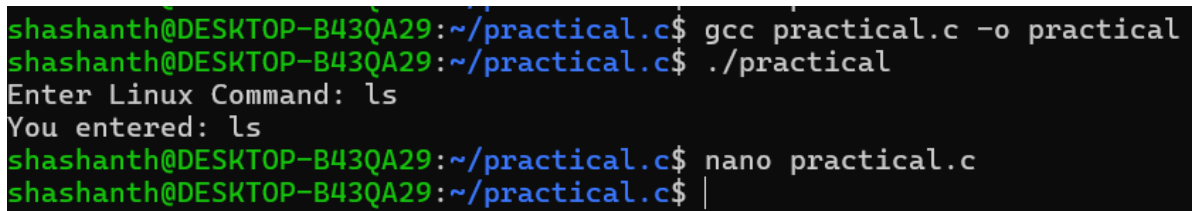
```
shashanth@DESKTOP-B43QA: × + v
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    char cmd[50];

    printf("Enter Linux Command: ");
    scanf("%s", cmd);

    printf("You entered: %s\n", cmd);

    return 0;
}
```

A screenshot of a terminal window showing the compilation and execution of the C program. The user runs 'gcc practical.c -o practical' and then './practical'. The program prompts for a command, the user enters 'ls', and the program outputs 'ls'. The user then runs 'nano practical.c' to edit the file.

```
shashanth@DESKTOP-B43QA29:~/practical.c$ gcc practical.c -o practical
shashanth@DESKTOP-B43QA29:~/practical.c$ ./practical
Enter Linux Command: ls
You entered: ls
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical.c
shashanth@DESKTOP-B43QA29:~/practical.c$ |
```

1.b

```
shashanth@DESKTOP-B43QA29: ~$ nano practical.c
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

```
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical.c
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical.c
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical1.c
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical1.c
shashanth@DESKTOP-B43QA29:~/practical.c$ gcc practical1.c -o practical1
shashanth@DESKTOP-B43QA29:~/practical.c$ ./practical1
Hello
Hello
shashanth@DESKTOP-B43QA29:~/practical.c$ |
```

1.c

```
shashanth@DESKTOP-B43QA: x + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child executing ls...\n");
        execl("/bin/ls", "ls", "-l", NULL);
    }
    else
    {
        printf("Parent Process\n");
    }

    return 0;
}
```

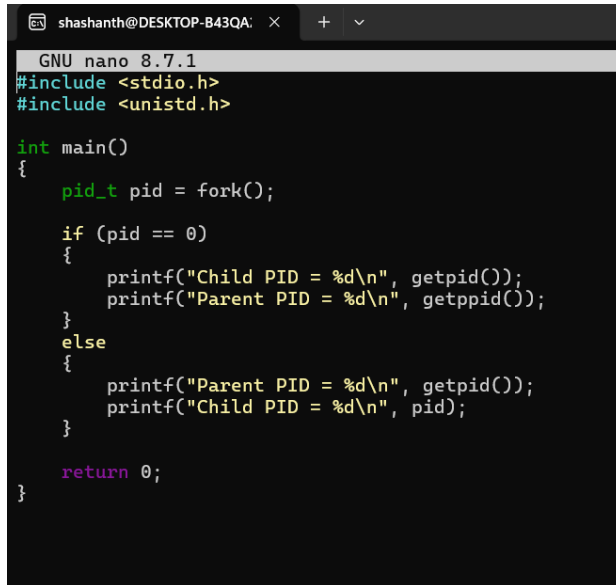
```
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical2.c
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical2.c
shashanth@DESKTOP-B43QA29:~/practical.c$ gcc practical2.c -o practical2
shashanth@DESKTOP-B43QA29:~/practical.c$ ./practical2
Parent Process
Child executing ls...
shashanth@DESKTOP-B43QA29:~/practical.c$ total 60
-rwxr-xr-x 1 shashanth shashanth 16056 Aug  5 03:52 practical
-rw-r--r-- 1 shashanth shashanth  167 Aug  5 03:56 practical.c
-rwxr-xr-x 1 shashanth shashanth 16000 Aug  5 03:58 practical1
-rw-r--r-- 1 shashanth shashanth  106 Aug  5 03:58 practical1.c
-rwxr-xr-x 1 shashanth shashanth 16040 Aug  5 04:02 practical2
-rw-r--r-- 1 shashanth shashanth  269 Aug  5 04:02 practical2.c
```

1.d

```
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical2.c
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical2.c
shashanth@DESKTOP-B43QA29:~/practical.c$ gcc practical2.c -o practical2
shashanth@DESKTOP-B43QA29:~/practical.c$ ./practical2
Parent Process
Child executing ls...
shashanth@DESKTOP-B43QA29:~/practical.c$ total 60
-rwxr-xr-x 1 shashanth shashanth 16056 Aug  5 03:52 practical
-rw-r--r-- 1 shashanth shashanth  167 Aug  5 03:56 practical.c
-rwxr-xr-x 1 shashanth shashanth 16000 Aug  5 03:58 practical1
-rw-r--r-- 1 shashanth shashanth  106 Aug  5 03:58 practical1.c
-rwxr-xr-x 1 shashanth shashanth 16040 Aug  5 04:02 practical2
-rw-r--r-- 1 shashanth shashanth  269 Aug  5 04:02 practical2.c
```

```
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical3.c
shashanth@DESKTOP-B43QA29:~/practical.c$ gcc practical3.c -o practical3
shashanth@DESKTOP-B43QA29:~/practical.c$ ./practical3
Child Executing
Parent Executing
shashanth@DESKTOP-B43QA29:~/practical.c$ |
```

1.e



```
shashanth@DESKTOP-B43QA29: x + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
    {
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```

```
shashanth@DESKTOP-B43QA29:~/practical.c$ nano practical4.c
shashanth@DESKTOP-B43QA29:~/practical.c$ gcc practical4.c -o practical4
shashanth@DESKTOP-B43QA29:~/practical.c$ ./practical4
Parent PID = 1040
Child PID = 1041
Child PID = 1041
Parent PID = 590
shashanth@DESKTOP-B43QA29:~/practical.c$ |
```